

EPPLEY FIELD, NE, USA

WMO: 725500

Lat: 41.310N

Lon: 95.899W

Elev: 299

StdP: 97.78

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14942

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.3	-16.9	-23.8	0.5	-18.6	-21.5	0.6	-16.3	14.3	-6.0	12.6	-6.3	4.0	350	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	35.0	24.5	33.4	24.1	31.8	23.3	26.4	32.2	25.4	31.1	24.5	29.9	6.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.4	30.3	23.7	19.2	29.2	22.8	18.1	28.1	83.9	32.2	79.5	31.1	75.6	30.0	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.8	9.3	DB	-23.5	37.7	2.6	2.0	-25.4	39.1	-27.0	40.3	-28.4	41.4	-30.3	42.9	(4)
(5)			WB	-23.8	27.8	2.6	0.8	-25.7	28.4	-27.2	28.8	-28.6	29.3	-30.5	29.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-4.6	-2.2	4.6	11.2	17.3	23.0	25.4	24.0	19.7	12.3	4.6	-2.2	(6)	
	DBStd	11.72	6.73	6.80	6.77	5.54	4.76	3.74	3.33	3.15	4.60	5.27	5.61	6.10	(7)	
	HDD10.0	1633	451	344	192	50	3	0	0	0	1	36	177	380	(8)	
	HDD18.3	3304	709	576	427	221	78	7	1	2	38	196	412	638	(9)	
	CDD10.0	2056	0	2	25	87	229	389	476	433	292	108	15	1	(10)	
	CDD18.3	685	0	0	1	8	46	146	218	177	79	10	0	0	(11)	
	CDH23.3	6847	0	0	17	109	472	1461	2332	1609	744	100	3	0	(12)	
	CDH26.7	2743	0	0	2	30	156	583	1047	638	262	24	0	0	(13)	
(14) Wind	WSAvg	4.5	4.6	4.7	5.1	5.3	4.7	4.3	3.8	3.6	4.1	4.4	4.6	4.4	(14)	
(15) Precipitation	PrecAvg	842	23	28	52	86	124	131	100	120	63	53	33	26	(15)	
	PrecMax	1096	43	64	114	188	234	286	244	262	169	132	69	122	(16)	
	PrecMin	567	8	7	6	12	43	18	8	32	10	0	0	0	(17)	
	PrecStd	142	10	14	28	37	53	68	56	66	43	37	22	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	18.6	25.5	30.4	33.5	35.5	38.3	36.2	34.1	29.8	22.9	14.9	(19)	
		MCWB	6.6	10.1	14.7	17.8	21.6	23.7	24.4	24.4	22.7	19.7	14.4	9.9	(20)	
	2%	DB	9.7	14.0	21.9	26.5	30.4	33.5	35.5	34.0	31.9	26.1	19.1	11.3	(21)	
		MCWB	5.2	8.6	13.7	16.4	20.5	23.6	25.2	24.5	22.4	17.7	12.8	7.2	(22)	
	5%	DB	6.8	10.3	18.5	23.5	28.2	31.8	33.8	31.9	29.7	23.4	15.9	8.6	(23)	
		MCWB	3.5	5.9	12.3	14.6	18.8	22.8	24.9	24.0	21.4	16.0	10.8	5.2	(24)	
	10%	DB	4.2	7.1	15.2	20.8	26.0	30.1	31.8	30.2	27.6	20.9	13.3	6.0	(25)	
		MCWB	1.9	3.9	10.0	13.3	18.1	22.1	24.1	23.1	20.4	14.9	8.9	3.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	12.7	17.3	19.4	23.4	26.3	27.9	27.2	24.9	21.7	16.0	11.9	(27)	
		MCDB	12.2	15.8	22.5	27.1	29.6	32.1	33.7	32.9	31.6	27.2	20.2	13.9	(28)	
	2%	WB	5.6	8.8	15.3	17.5	21.8	25.0	26.8	26.0	23.6	19.3	13.7	7.7	(29)	
		MCDB	9.5	13.6	19.2	24.4	28.4	31.3	32.8	31.5	29.4	23.6	17.7	10.4	(30)	
	5%	WB	3.7	6.1	12.9	15.8	20.4	24.0	25.7	25.0	22.6	17.6	11.5	5.3	(31)	
		MCDB	6.4	9.9	17.9	22.1	26.3	30.0	31.5	30.2	28.0	21.8	15.5	8.3	(32)	
	10%	WB	1.9	4.0	10.1	14.1	19.0	23.0	24.9	24.0	21.6	15.6	9.0	3.5	(33)	
		MCDB	4.1	7.0	15.3	19.8	24.4	28.9	30.4	28.7	26.4	20.1	13.0	5.9	(34)	
(35) Mean Daily Temperature Range	MDBR		9.7	10.1	11.6	12.0	11.5	10.9	10.6	10.5	12.0	11.7	10.7	9.2	(35)	
	5% DB	MCDBR	13.8	15.0	16.8	16.7	14.4	12.4	12.2	12.3	13.6	14.8	15.1	13.3	(36)	
		MCWBR	9.9	9.8	9.6	8.8	6.4	5.3	5.0	5.3	6.1	7.7	9.1	9.6	(37)	
	5% WB	MCDBR	12.9	14.2	14.6	14.2	12.3	11.0	10.8	10.8	11.6	12.4	13.5	12.2	(38)	
		MCWBR	9.5	9.7	8.8	8.4	6.2	5.8	5.5	5.5	6.1	7.6	9.1	9.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.283	0.318	0.332	0.388	0.412	0.417	0.429	0.418	0.378	0.335	0.300	0.286	(40)	
	taud		2.445	2.365	2.390	2.253	2.255	2.300	2.260	2.299	2.366	2.473	2.498	2.450	(41)	
	Ebn at Noon		879	890	917	882	866	860	844	844	856	862	852	842	(42)	
	Edh at Noon		77	97	107	132	135	129	133	125	109	87	73	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.00	2.94	3.81	4.87	5.79	6.43	6.49	5.60	4.62	3.23	2.24	1.68	(44)	
	RadStd		0.18	0.17	0.34	0.35	0.44	0.50	0.43	0.37	0.30	0.38	0.16	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	+70
Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page